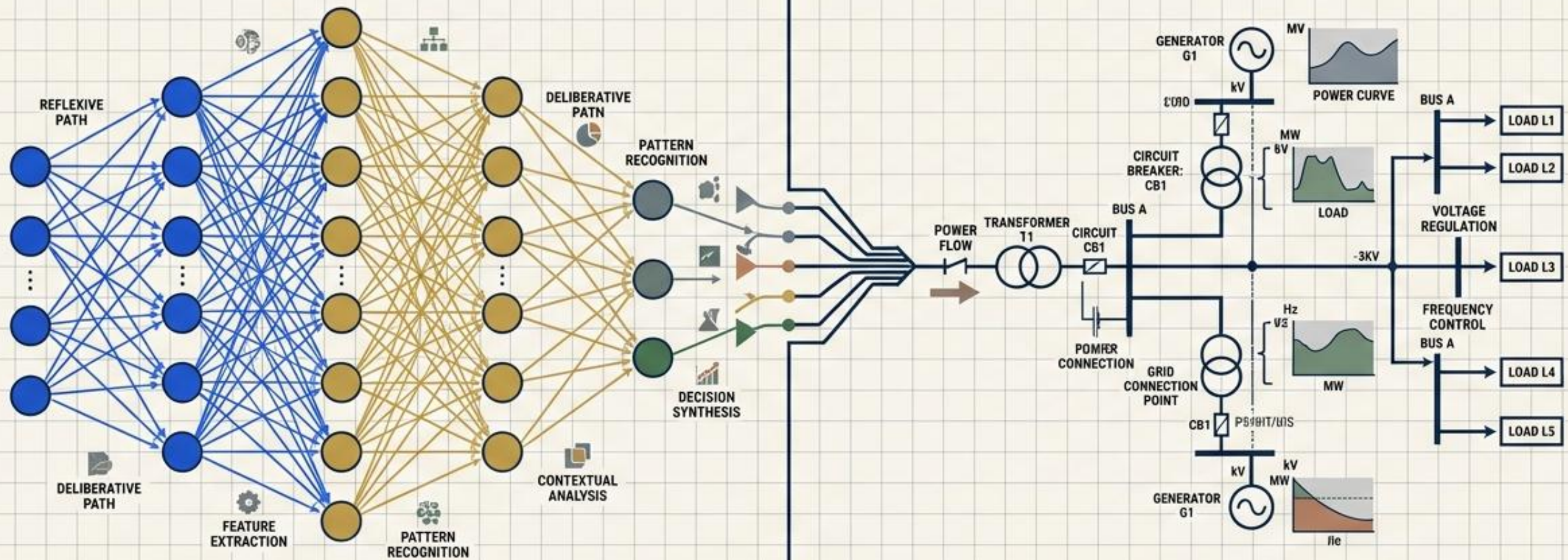


Cognitive Adaptive Power System Management

A Brain-Inspired Dual-Process AI Framework Validated
Through Controller-Hardware-in-the-Loop Testing



PhD Defense Presentation | Mahmoud Kiasari | Dalhousie University

Comprehensive Validation Yields Unprecedented Grid Performance

System 1 (Reflexive)

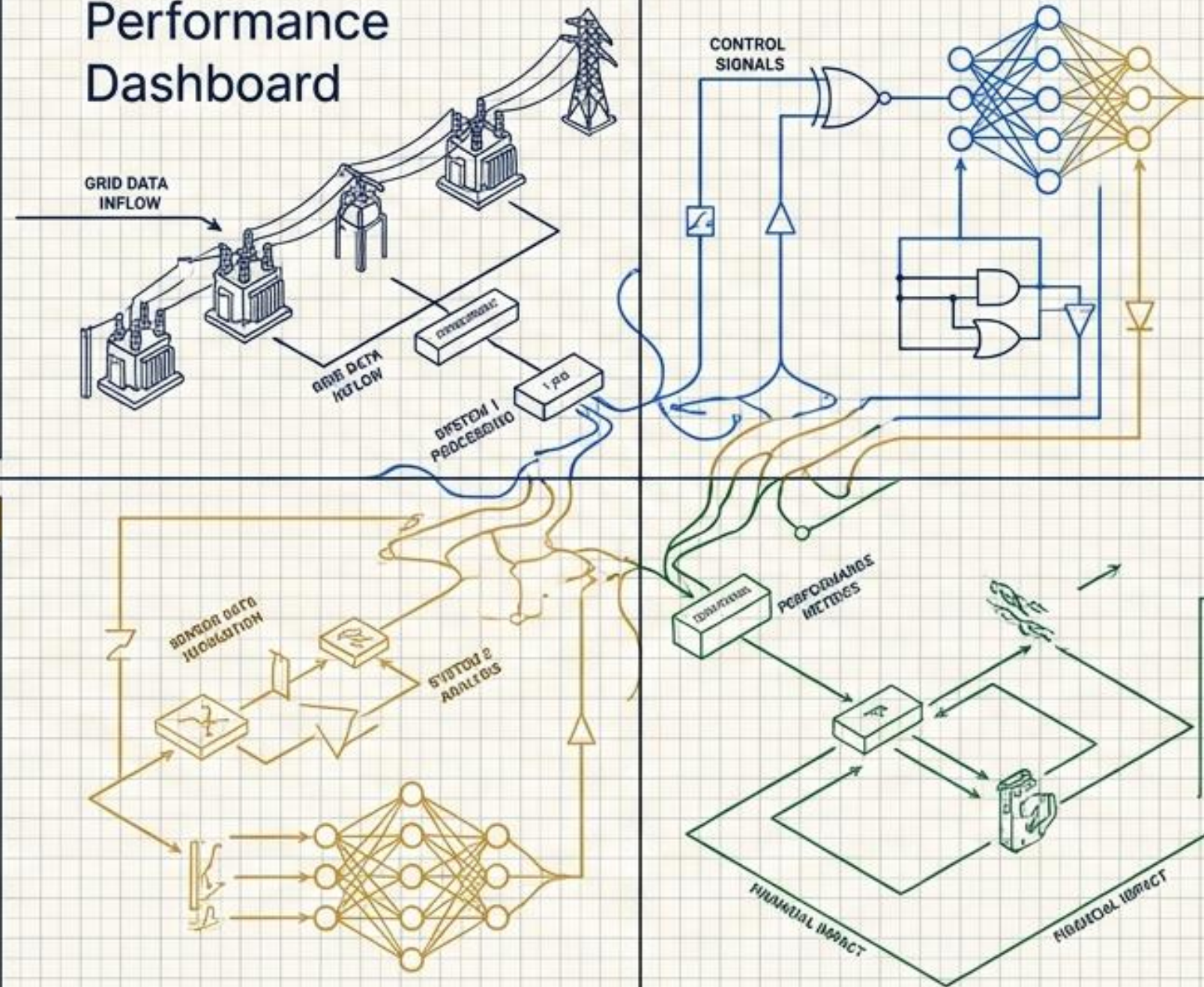
37%



Faster Response

Sub-10ms response times to critical events achieved on physical HIL.

Performance Dashboard



System 1

System 2

42%



Greater Stability

Overall voltage stability margin increased during extreme grid events.

System 2 (Deliberative)

95.8%



Detection Accuracy

Fault detection and localization accuracy via multi-modal fusion.

12.4%



Cost Savings

Annual operational cost reduction, yielding \$51.5M/year in savings.

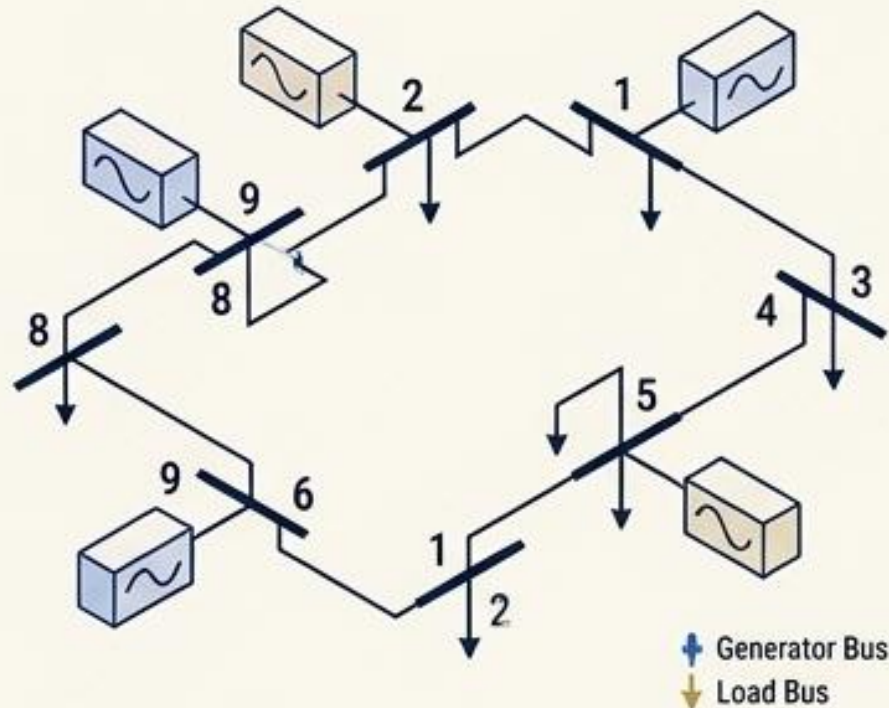
Performance validated consistently from Python simulations to real-time hardware.

Progressive Scaling From Debug to Dense Grid Topologies

Validation progression ensures robustness from simulation to real-time hardware environments.

Stage 1: Offline Python Validation & Algorithm Training

OpenAI Gym-like PYPOWER interface; rapid iteration without real-time constraints.

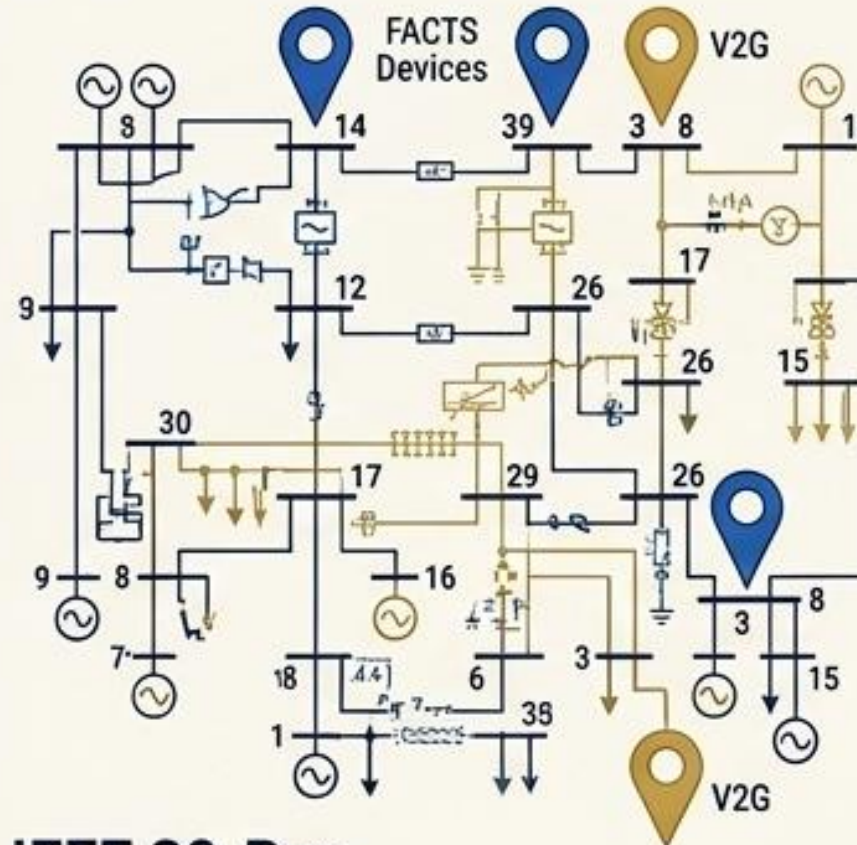


IEEE 9-Bus

OpenAI Gym-like PYPOWER interface; rapid iteration without real-time constraints.

Stage 2: Primary Test System

High-fidelity Simulink environment; handles dynamic 20ms resolution load profiles.

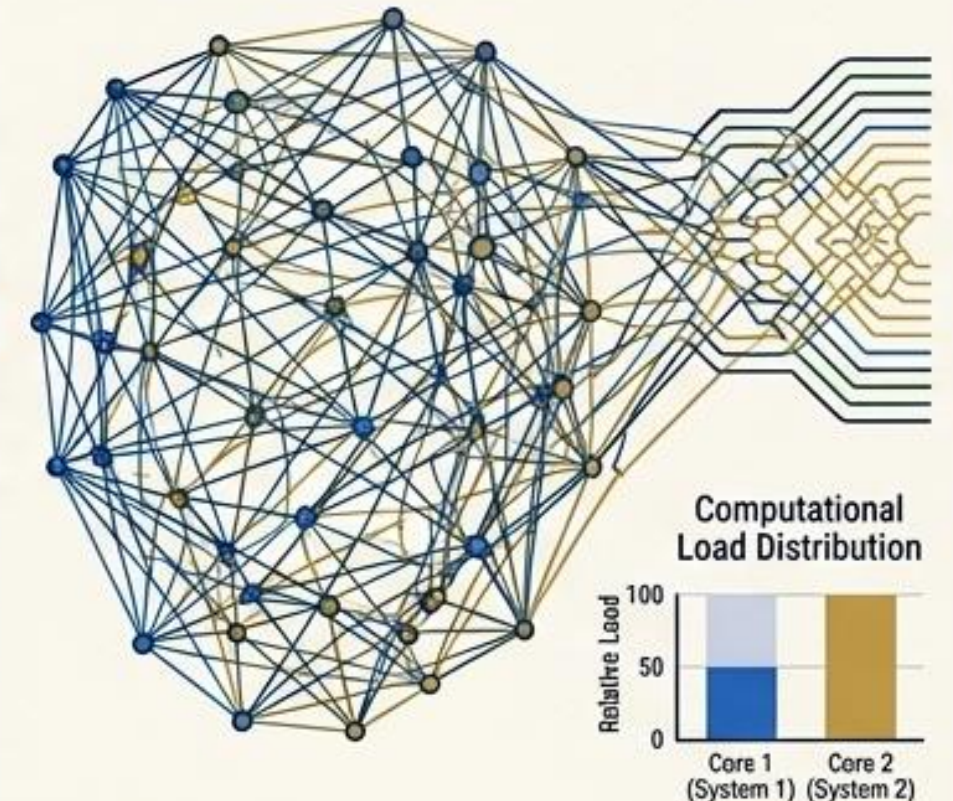


IEEE 39-Bus

High-fidelity Simulink environment; handles dynamic 20ms resolution load profiles.

Stage 3: Scalability Validation

Validates multi-core allocation feasibility; achieves \$51.5M/year operational savings on larger topologies.



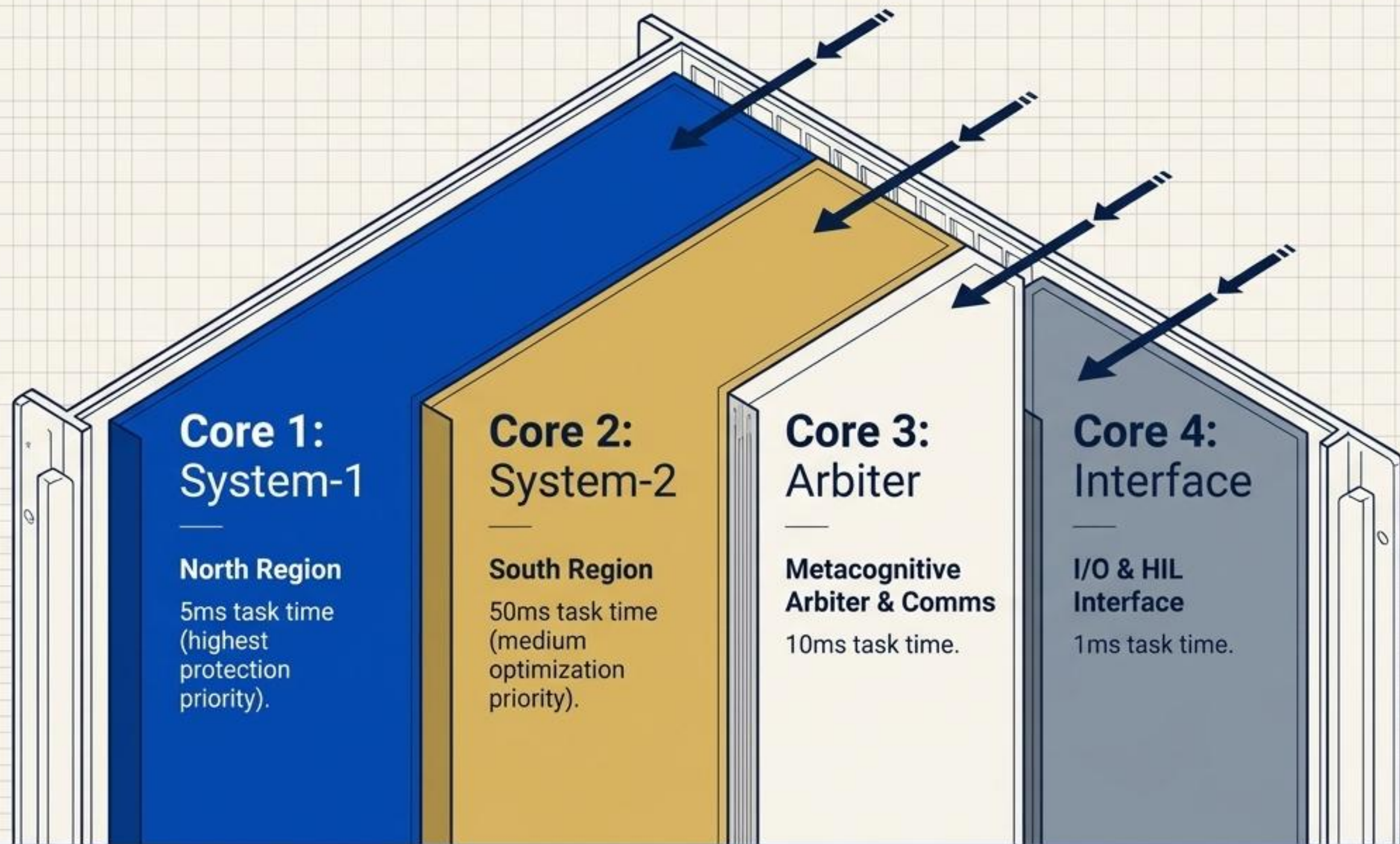
IEEE 118-Bus

Validates multi-core allocation feasibility; achieves \$51.5M/year operational savings on larger topologies.

Savings based on annual operational cost reduction.

Performance and scalability metrics validated through rigorous simulation and analysis across all stages.

Bridging Theory and Practice on OPAL-RT Real-Time Hardware

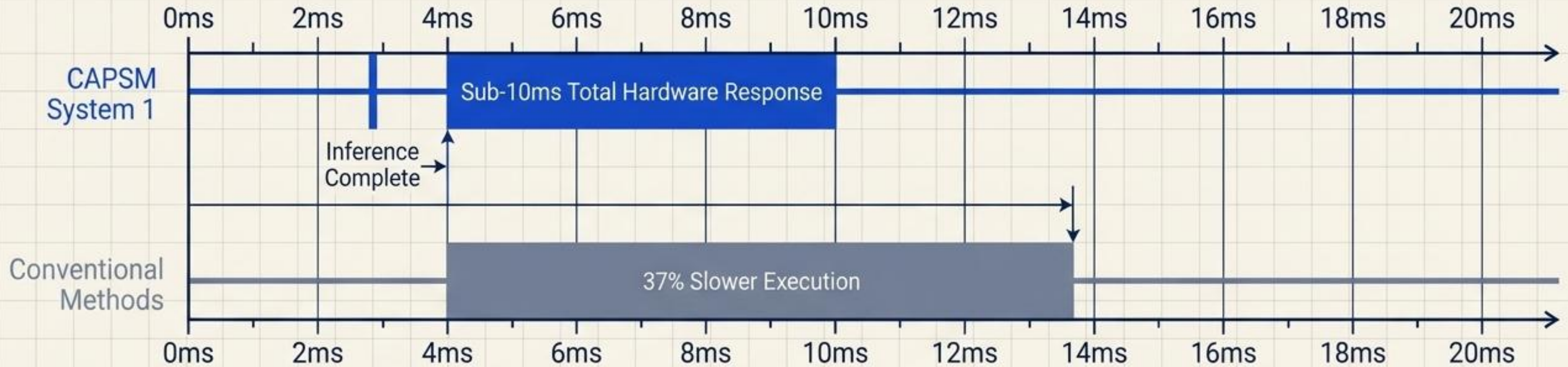


Hardware Reality

Executes at a 50 μ s Electromagnetic Transient (EMT) time step.

AI runs directly on simulator CPUs.

The Reflexive Brain Executes Sub-10ms Protection Responses



Mechanism

CNN extracts spatial patterns from PMU bus data; LSTM tracks temporal oscillation dynamics.

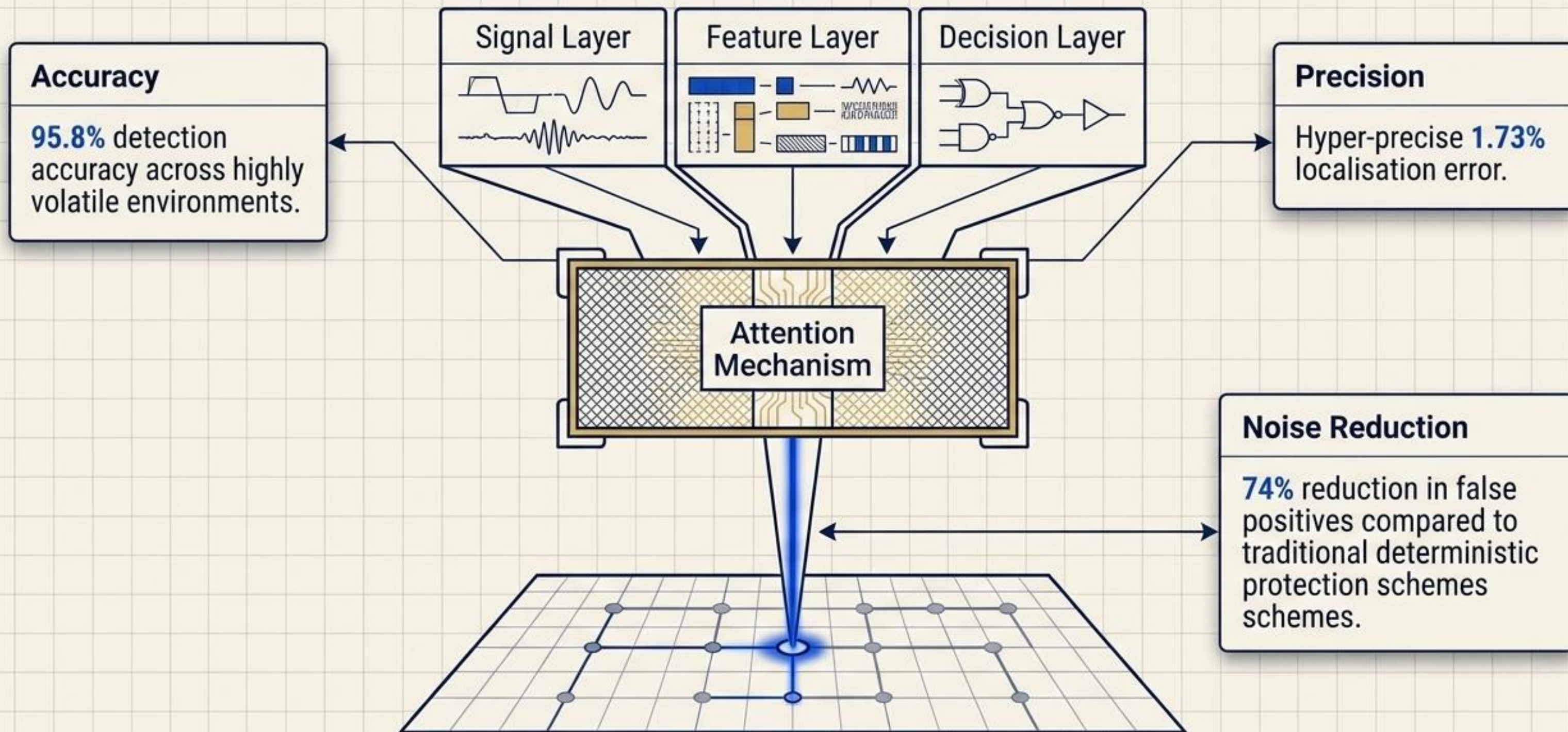
Hardware Reality

Achieves sub-10ms fault detection on physical HIL hardware, not just theoretical simulation.

Metacognitive Trigger

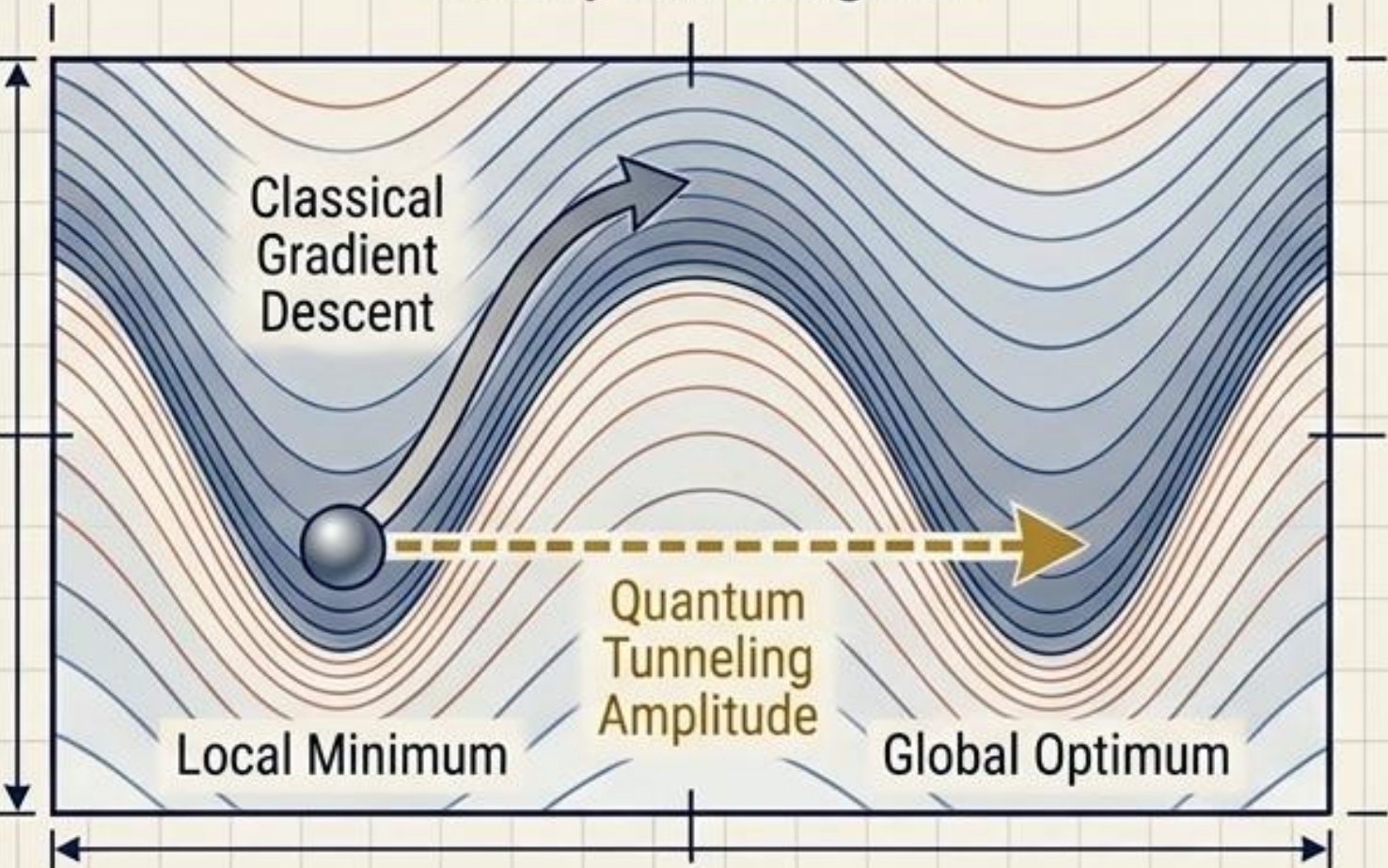
Automatically activated during high-urgency, low-novelty events (e.g., sudden voltage drops).

Multi-Modal Fusion Achieves Near-Perfect Fault Localisation

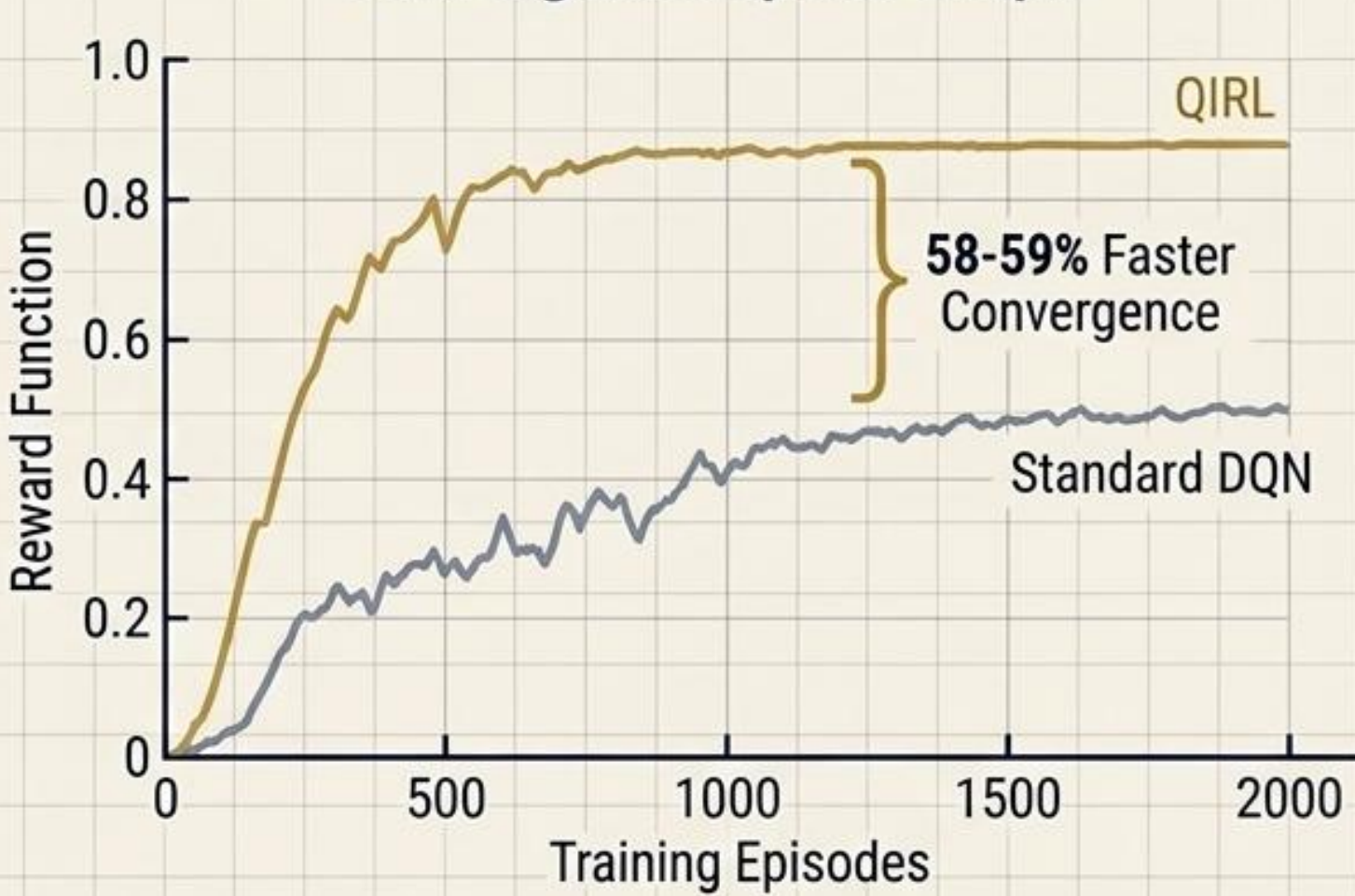


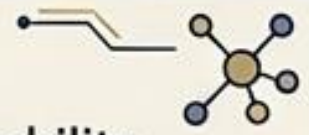
Quantum-Inspired Reinforcement Learning Tunnels Out of Local Optima

Conceptual Diagram

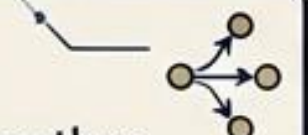


Convergence Speed Graph

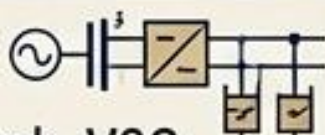


The Mechanism 

Uses complex-valued probability amplitudes to evaluate multiple states simultaneously (superposition).

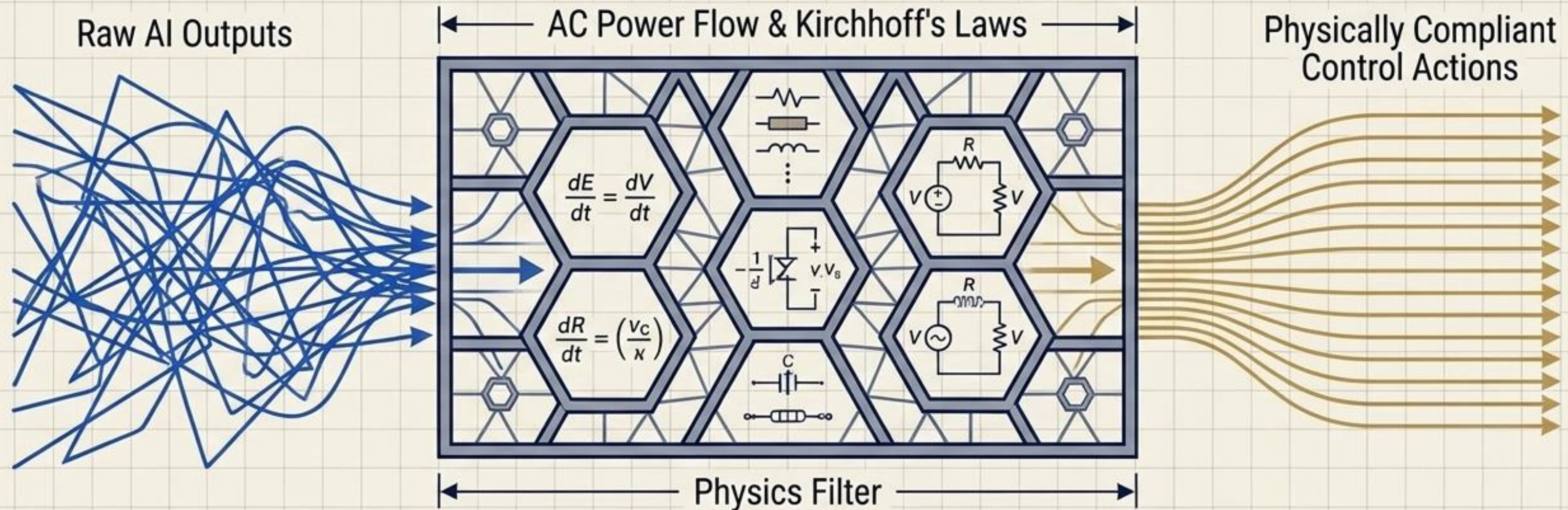
The Result 

Converges 58% to 59% faster than standard Deep Q-Networks (DQN).

The Application 

Handles economic dispatch, V2G scheduling, and coordinated FACTS control in 50ms intervals.

Physics-Informed Neural Networks Guarantee Grid Safety



Embedded Physics

Physical constraints are mathematically embedded directly into the training loss function.

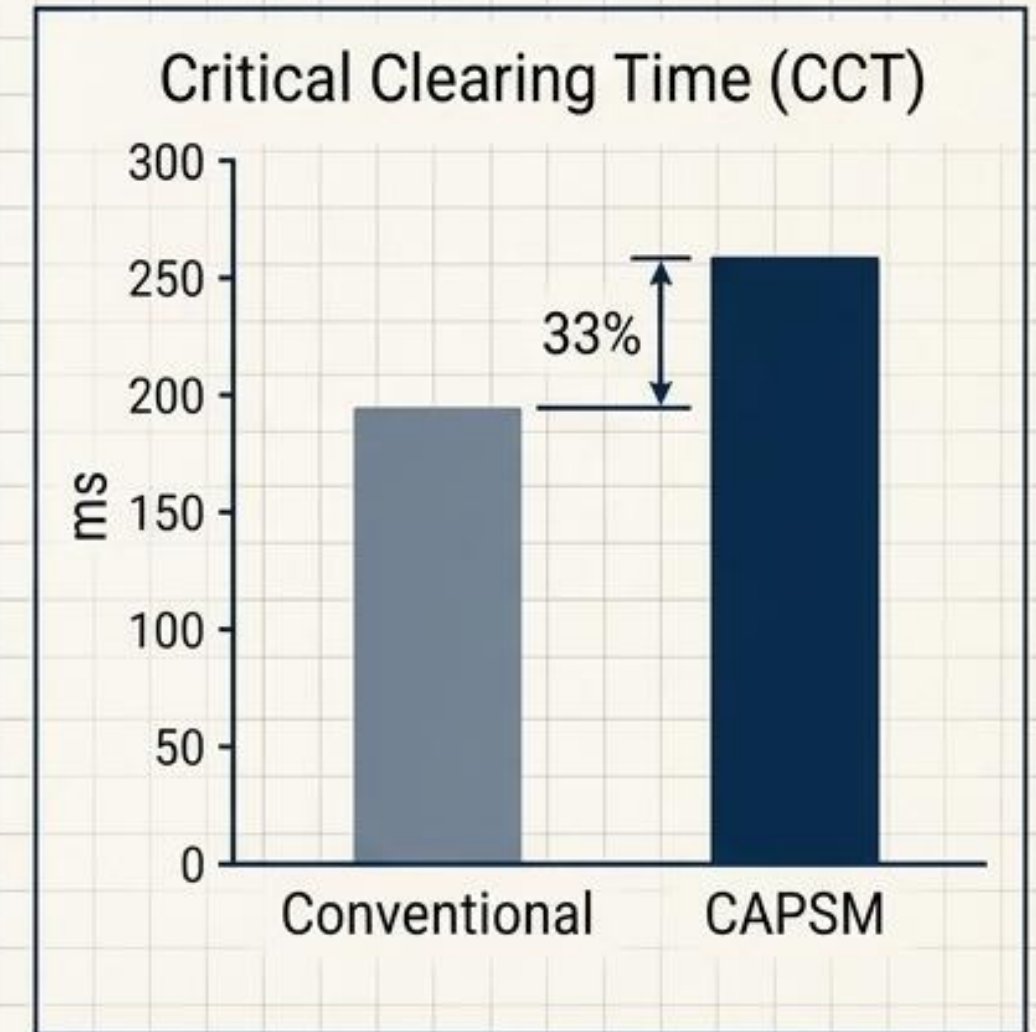
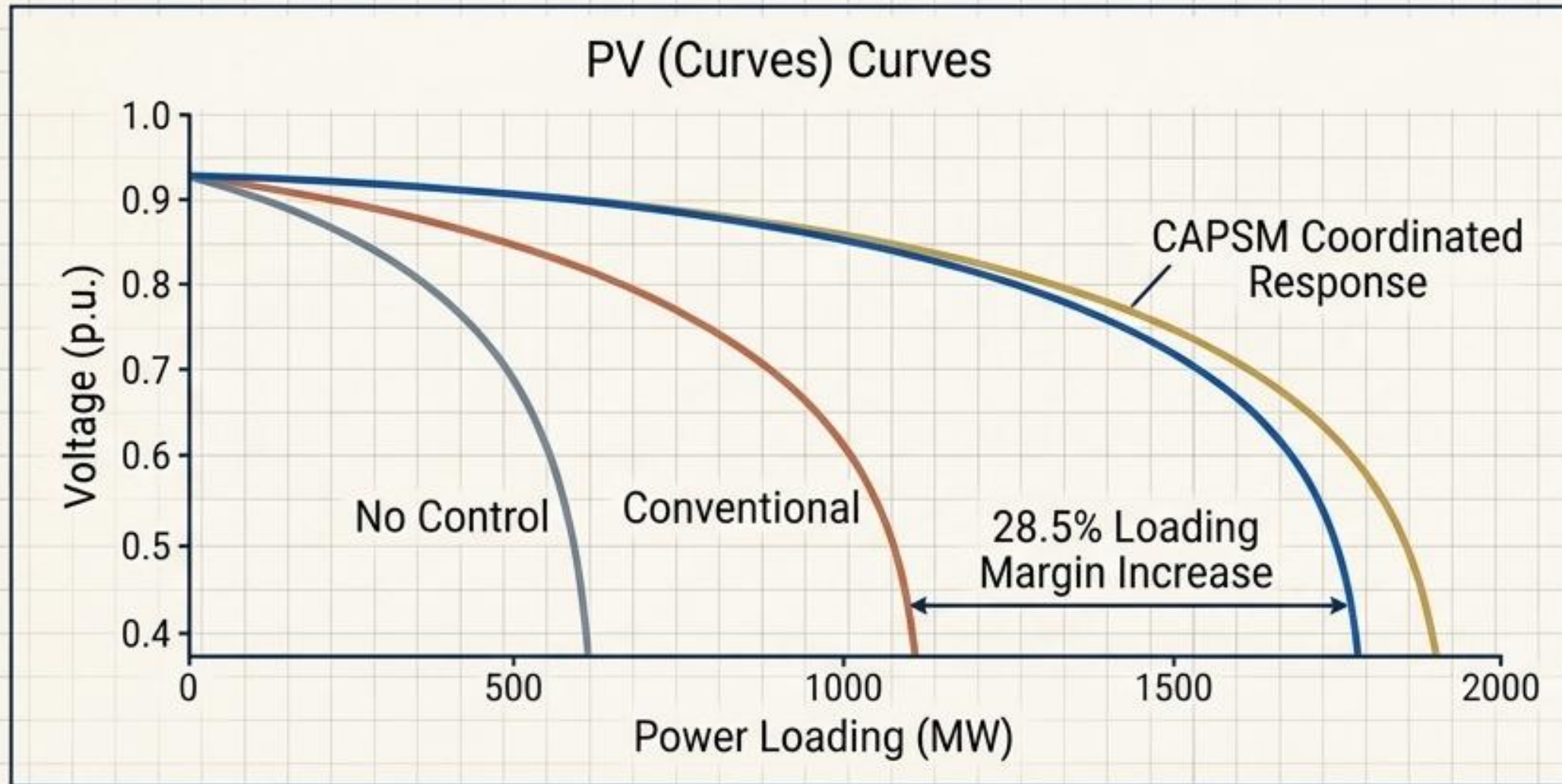
Constraint Compliance

99.97% physics constraint satisfaction rate achieved.

Violation Reduction

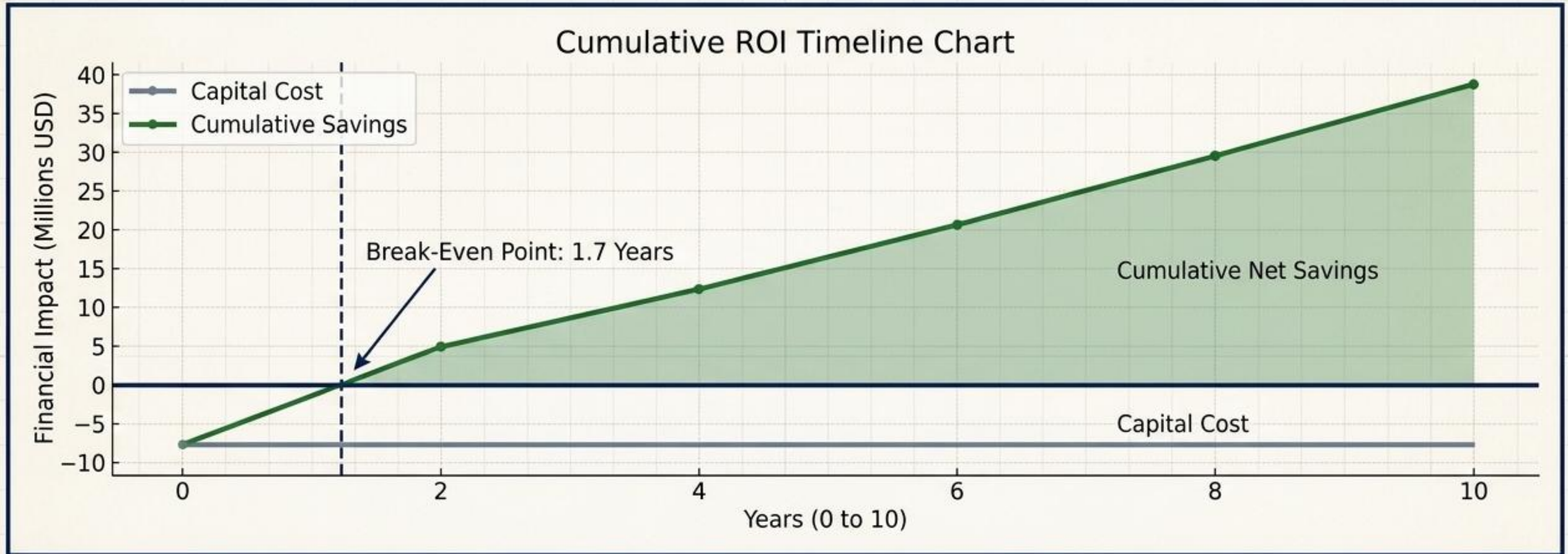
Erroneous control actions dropped from 3,130 to just **11** in 10,000 operational cycles.

Coordinated Control Radically Expands Voltage Stability Margins



- Voltage Stability Coordinated response yields a 42% better overall voltage stability margin.
- Loading Margin 28.5% increase in loading capacity before collapse.
- Transient Stability 20-33% improvement in Critical Clearing Time during severe grid faults.

Bidirectional EV Integration Drives Immediate Economic Returns



V2G Coordination

Synchronizes Vehicle-to-Grid charging to shave peaks and provide frequency regulation.

Operational Savings

Drives a 12.4% total operational cost reduction.

Financial Metrics

Achieves a 1.7-year break-even ROI and an 83.7% Internal Rate of Return (IRR).

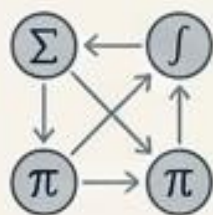
CAPSM Outperforms All Conventional and State-of-the-Art Methods

	Response Time	Adaptability to Uncertainty	Physics Constraint Compliance	Real-Time HIL Readiness
Conventional SCADA				
Rule-Based Expert Systems				
Standard Deep RL (DRL)				
CAPSM Framework				

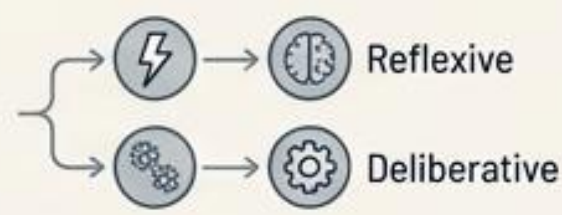
Takeaway: CAPSM is the only solution that bridges theoretical AI performance with fully verified HIL real-time physical constraints.

Eight Core Contributions Advance the Field of Cognitive Grids

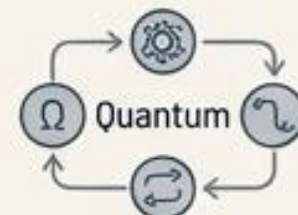
C1: Unified Math Framework



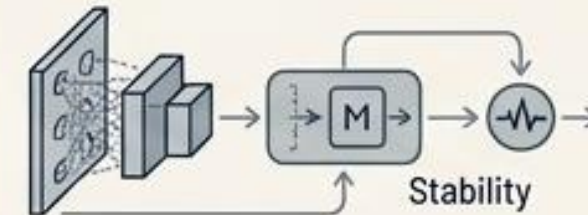
C2: Dual-Process Architecture



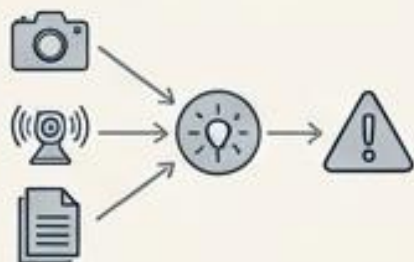
C3: Quantum-Inspired RL (QIRL)



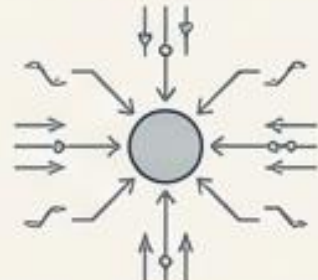
C4: CNN-LSTM Stability Model



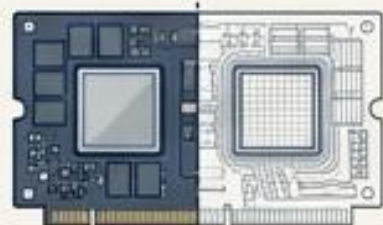
C5: Multi-Modal Fault Detection



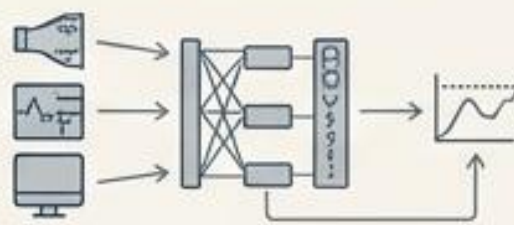
C6: Coordinated Control



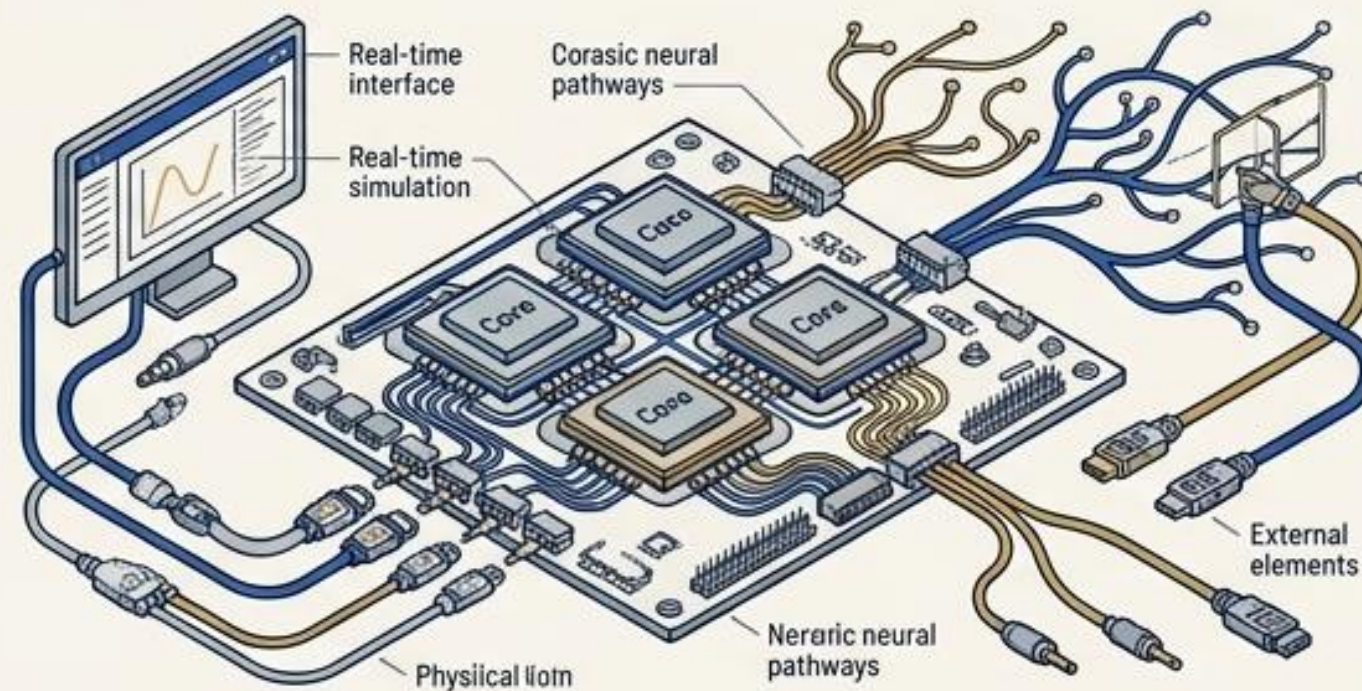
C7: Digital Twin Methodology



C4: CNN-LSTM Stability Model



C8: Hardware-in-the-Loop Validation on 4-Core OPAL-RT



The critical bridge from simulation to practical implementation, validating real-time constraint compliance and sub-10ms response times.

Dissemination

Journals

Conferences

Open Source Data

Transparent Acknowledgment of Current Limitations

Known Constraints

Physical & HIL Constraints

- Current OPAL-RT validation is limited to selected components.
- EV fleet models require further unaggregated physical testing to validate complex localized thermal dynamics.

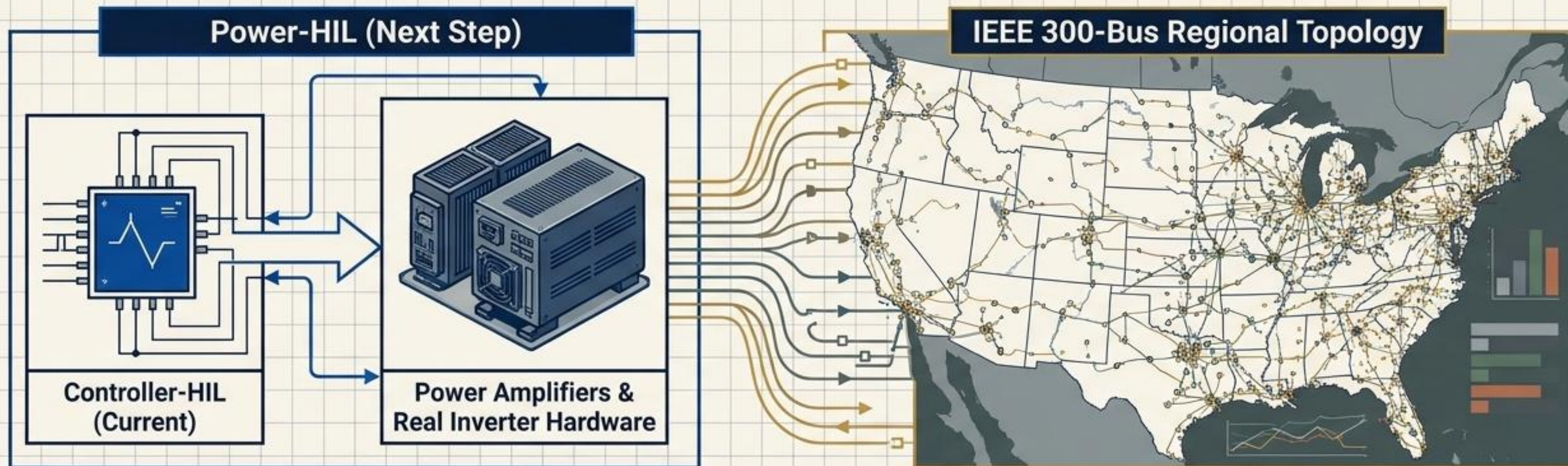
Computational Bottlenecks

- Scaling ePHASORSIM integration for 1000+ node EMT constraints remains processor-bound on current 4-core hardware architectures.

Regulatory Barriers

- Autonomous AI dispatch without human-in-the-loop oversight faces strict, current grid code and liability barriers.
- Certification pathways for non-deterministic AI controllers do not yet exist.

The Horizon Extends to Power-HIL and Regional Grid Deployment



Immediate Next Step

- Transition from testing control algorithms to testing closed-loop physical inverter interactions.

Scaling Up

- Bringing the IEEE 300-bus system into real-time OPAL-RT simulation.

Industry Roadmap

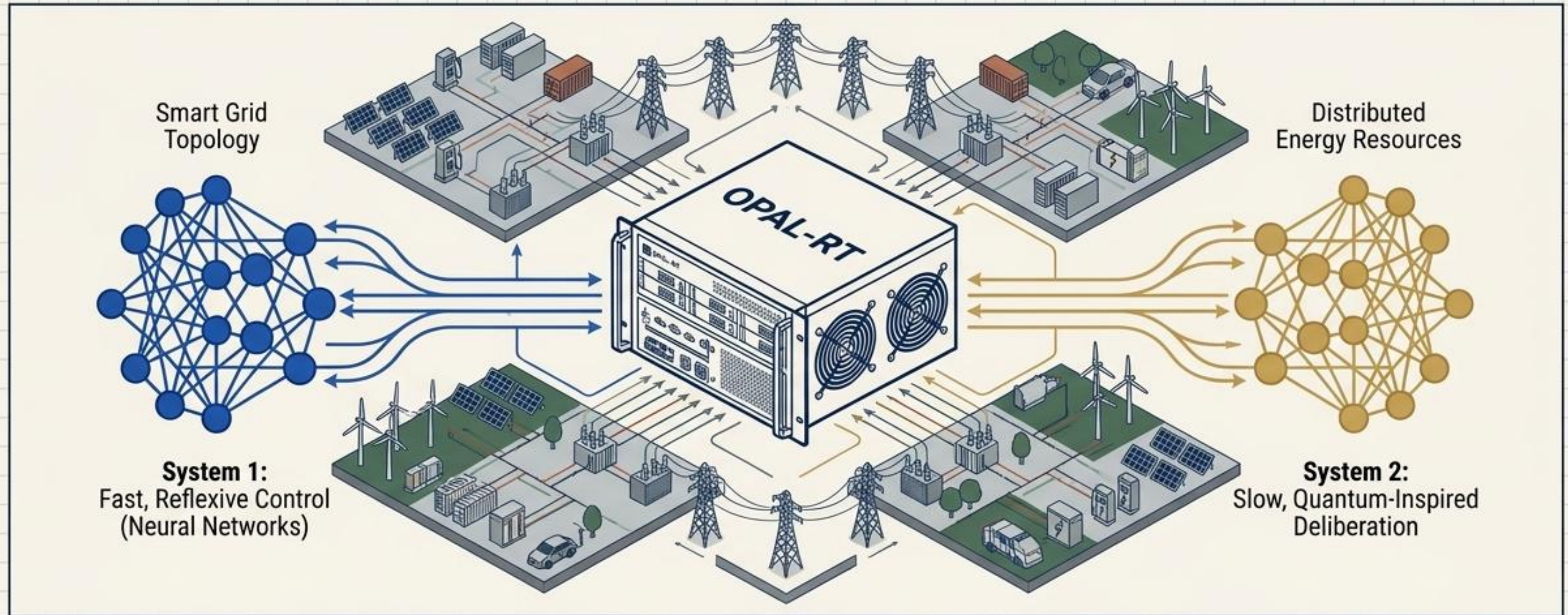
Phase 1:
Advisory Mode

Phase 2:
Limited Closed-Loop

Phase 3:
Integrated Control

Phase 4:
Advanced Autonomous

A Validated Blueprint for Autonomous Energy Communities



Timeline of Control: Shifting from Centralized/Deterministic to Distributed/Cognitive

CAPSM successfully bridges the theoretical AI-to-practice gap.

By combining the speed of reflexive neural networks with the depth of quantum-inspired deliberation, we provide a fast, safe, and economically viable brain for the modern renewable grid.

Final Thesis Statement:
The journey toward truly intelligent power systems—validated on real hardware—has officially begun.